

Japan Travel Itinerary Planner with AI Integration

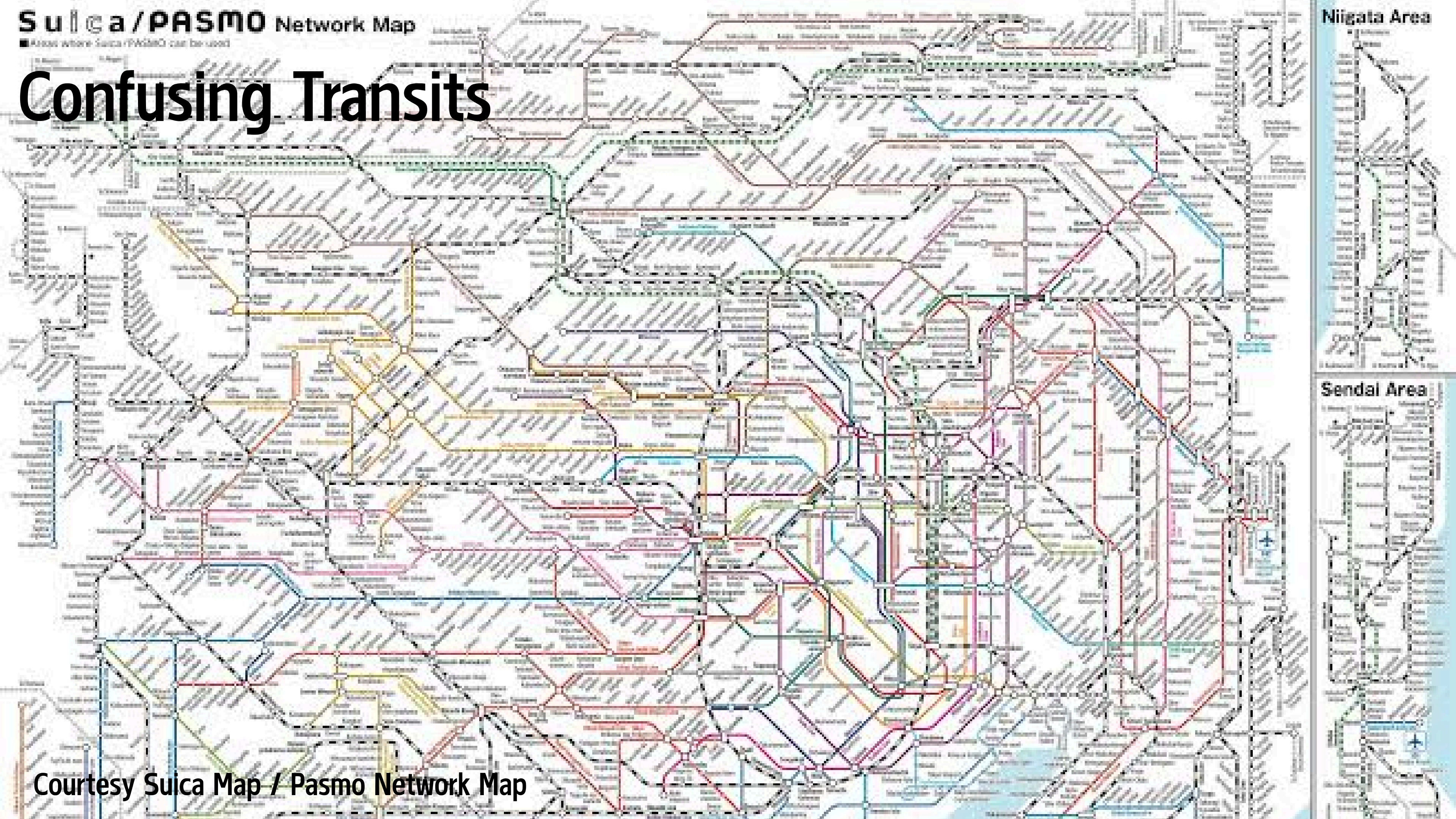
PE6203 Generative AI Solution



By Group 1



Confusing Transits





Confusing Diet Restrictions

Travelers must solve navigation and dietary constraints together

Navigation Confusion

30+ travel passes

with different coverage, operators and validity periods

45%

struggled to find the correct route

38%

struggled to locate where to board or get off

Dietary Restrictions

86%

of surveyed Muslim travelers worried about finding halal food

53%

of surveyed vegan travelers would avoid restaurants without suitable meal options

Sources:

Japan Tourism Agency, FY2025 Survey on the Acceptance Environment for Foreign Visitors, p.18 (2026); JNTO, Regional Rail Passes in Japan (accessed September 2026); Halal Navi, Survey of 1,581 Muslims (2024); Japan Tourism Agency, Guide for Accommodating Vegetarian and Vegan Travelers, p.10 (2020)



SCAN HERE

Wayfinder Japan

A personalised Japan itinerary AI with transparent checks and rail-pass cost comparisons

Smart Navigation

Efficient travel with fare and pass comparison

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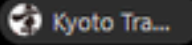
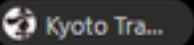
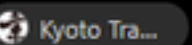
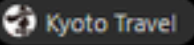
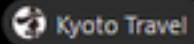
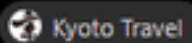
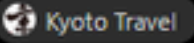
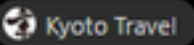
Suitable Meals

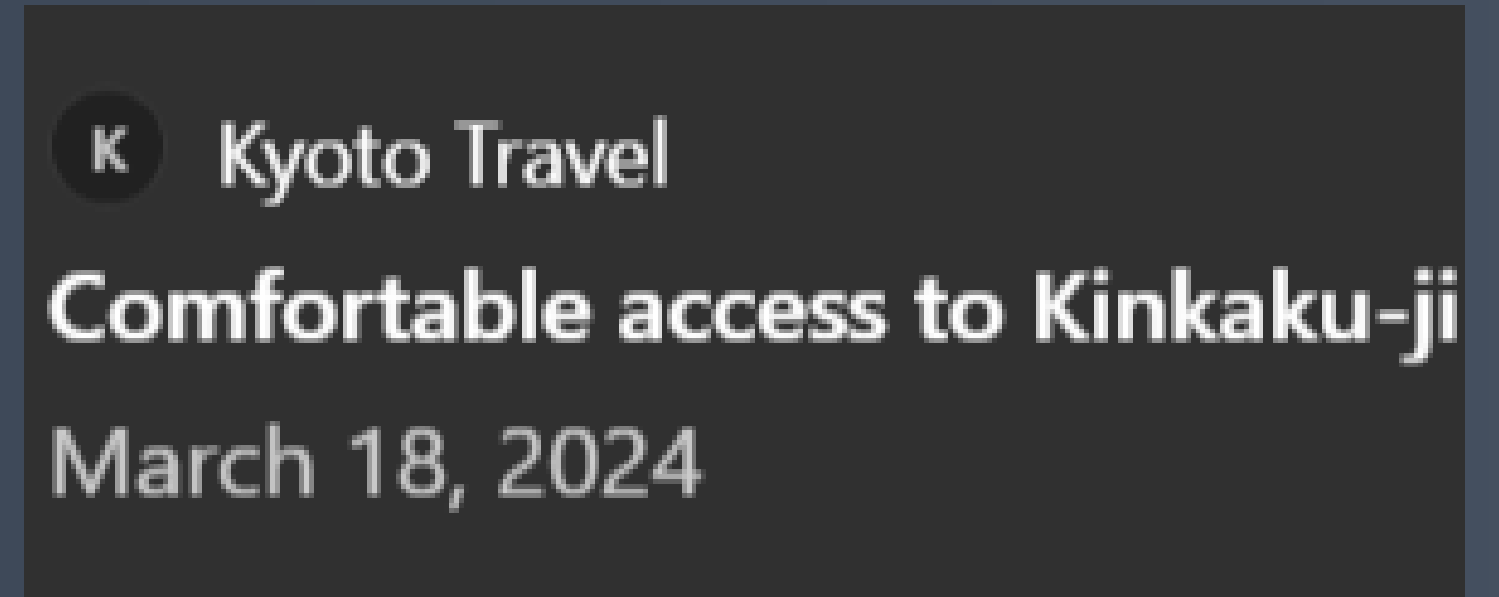
Halal and vegan options near each planned stop

Navigation and meals planned together



Why not ChatGPT?

Information in my itinerary	Source
General recommendation to combine trains, subway, buses and walking	Kyoto Travel — Getting Around Kyoto 
Kyoto → Arashiyama by JR Sagano Line	Kyoto Travel — Comfortable access to Saga/Arashiyama 
JR Sagano Line takes about 11 min by rapid train to Saga-Arashiyama	Same official Kyoto Travel page 
Kyoto → Kinkaku-ji using subway + bus	Kyoto Travel — Comfortable access to Kinkaku-ji 
JR Enmachi + bus as an alternative for Kinkaku-ji	Same official page 
Kyoto → Kiyomizu/Gion using JR Nara Line → Tofukuji → Keihan	Kyoto Travel — Getting Around Kyoto 
Kiyomizu/Gion bus congestion	Kyoto Travel — Comfortable access to Higashiyama 
JR is useful for Fushimi Inari and Arashiyama	Kyoto Travel — Getting Around Kyoto by Subway & Train 

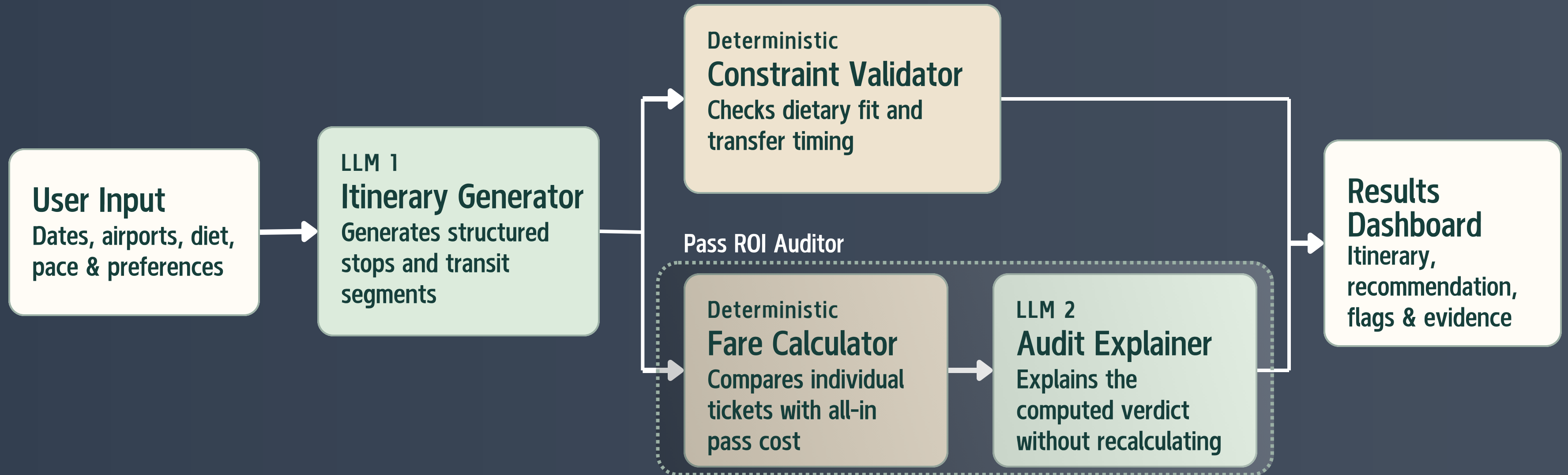


Outdated!

<https://chatgpt.com/c/6aa1d4d9-9528-83ec-bd88-60f8ab5611c7>

System Architecture

LLMs generate and explain; deterministic code checks and calculates.



Travel snippets

Keyword retrieval → Itinerary Generator

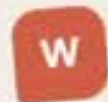
Dietary & travel-time tables

Direct lookup → Constraint Validator

Fare & pass tables

Direct lookup → Fare Calculator

Missing evidence stays unverified; missing required fares pause the cost verdict.



Wayfinder Japan

Plan with confidence

Evidence-led planning

A better route through Japan.

Shape the trip you want, then see the parts that hold up: fares, timing, dietary stops, and the evidence behind the answer.

Built for the decisions that matter

Suggestions come from the itinerary model. Cost and feasibility are checked separately, so a fluent answer never gets to mark its own homework.

Start with your trip

Enter your travel details. Add optional cities and preferences if you already know what you want, or leave them blank for a recommended route.

ARRIVAL DATE

01 / 10 / 2026

DEPARTURE DATE

07 / 10 / 2026

ARRIVAL TIME

10 :00

DEPARTURE TIME

18 :00

ARRIVAL AIRPORT

NGS — Nagasaki Airport, Nagasaki



DEPARTURE AIRPORT

KKX — Kikai Airport, Kagoshima



DIETARY REQUIREMENT

None



TRIP PACE

Relaxed



CITIES AND PREFERENCES (OPTIONAL)

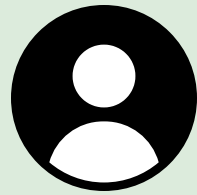
For example: Visit Tokyo and Kyoto, focus on historical areas, avoid nightlife, and travel mainly by train.

Testing

Criterion	Target	Actual	Met?	Notes
Financial Accuracy	≥90%	57.1% (8/14 scored)	✗	6 missing fare data
Dietary Adherence	100%	95.0% (19/20)	✗	Failure when HALAL and VEGAN
Geographic Plausibility	≥90%	95.0% (19/20)	✓	Bug that estimated 0 travel time for 10 min travel time
Faithfulness	≥90%	65.6% (410/625)	✗	Not every claim is cited

Error Mitigation Strategy

Key technical actions to boost accuracy and alignment



Expand Fare Coverage

Fill Data Gaps:

Populate missing Shinkansen supplement routes (e.g., Shin-Osaka ↔ Hiroshima).

Standardize Schema:

Enforce required fields across all rows



Audit Benchmark Data

Recalculate Baselines:

Manually verify ticket totals, pass pricing, and BUY/DO NOT BUY verdicts.

Update Test Cases: Patch test expectations in test cases to reflect accurate ground truth.



Improve Claim Alignment

Attach Source Dates:

Append timestamp metadata to all calculator claims.

- Fix Prompt Mismatches: Prevent AI verification drops caused by missing or mismatched time data.

QIA



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User intake form UI

Start with your trip

Enter your travel details. Add optional cities and preferences if you already know what you want, or leave them blank for a recommended route.

ARRIVAL DATE

01/10/2026



DEPARTURE DATE

07/10/2026



ARRIVAL TIME

10:00



DEPARTURE TIME

18:00



ARRIVAL AIRPORT

NRT — Narita International Airport, Chiba



DEPARTURE AIRPORT

KIX — Kansai International Airport, Osaka



DIETARY REQUIREMENT

Halal



TRIP PACE

Moderate



CITIES AND PREFERENCES (OPTIONAL)

For example: Visit Tokyo and Kyoto, focus on historical areas, avoid nightlife, and travel mainly by train.

Leave this blank if you want the planner to recommend the cities and complete route.

Generate & verify plan

Preview with sample

- Captures trip dates, flight times, airports, dietary needs, and travel pace.
- Accepts optional cities and free-text preferences to guide itinerary generation.
- Checks required fields and rejects inverted date ranges before generation.

Itinerary Generation & Results

Your plan and verification results

Live route · generated and checked locally

Live mode

● Verified = supported by curated evidence

● Flagged = conflicts with a hard check

● Unverified = insufficient curated evidence

PASS ROI AUDITOR

DO NOT BUY

Extra cost · ¥33,680

PASS SIDE

¥53,990

All-in cost

TICKETS

¥20,310

Individual fares

DIETARY CHECKS

6/ 6

Verified

GEOGRAPHIC CHECKS

13/ 14

Verified

Fare comparison uses curated airport and long-distance rail fares. Local subways, buses, trams, walking segments, and ordinary urban transfers are excluded unless a curated fare is available. Actual date-specific fares may vary.

Itinerary Generator

Day 1

3 stops

13:00

14:30

Senso-ji

Asakusa

Asakusa → Asakusa · Scheduled gap 15 min meets the required 10 min. Itinerary declares mode "Keisei Skyliner" but the table row is "walking"; the table is keyed by ward pair only, so this did not affect the verdict. · 2026-09-06

14:45

16:00

Sankyu Halal Japanese Food Asakusa

Asakusa · dining stop

Asakusa → Sumida · Scheduled gap 30 min meets the required 5 min. Itinerary declares mode "Local subway - Tokyo Metro Ginza Line" but the table row is "Tobu Skytree Line"; the table is keyed by ward pair only, so this did not affect the verdict. · 2026-09-07

16:30

18:30

Tokyo Skytree

Sumida

Day 2

4 stops

09:30

11:30

Meiji Jingu

Shibuya

Shibuya → Shibuya · Scheduled gap 30 min meets the required 10 min. Itinerary declares mode "Local train - JR

Audit notes

AUDITOR NARRATIVE

DO NOT BUY: point-to-point tickets save 33680 yen.

Point-to-point ticket total for this itinerary is 20310 yen, whereas the base price of "JR Pass 7-day Ordinary" is 50000 yen. In addition, Narita Airport Terminal 1 to Keisei-Ueno by Keisei Skyliner is not covered by "JR Pass 7-day Ordinary" and adds 2580 yen. Furthermore, Namba to Kansai Airport by Nankai Rapit Digital Ticket is not covered by "JR Pass 7-day Ordinary" and adds 1410 yen.

FARE BREAKDOWN

Narita Airport Terminal 1 → Keisei-Ueno

Keisei Skyliner

¥2,580

+ ¥2,580 not covered by pass

Tokyo → Kyoto

Hikari

¥13,650

Covered by pass

Kyoto → Shin-Osaka

Hikari

¥2,670

Covered by pass

Namba → Kansai Airport

Nankai Rapit Digital Ticket

¥1,410

+ ¥1,410 not covered by pass

CONSTRAINT VALIDATOR

Dietary

6 verified · 0 flagged · 0 unknown

Geographic

13 verified · 0 flagged · 1 unknown

- The itinerary model generates structured daily stops and transit segments.
- Results combine the proposed plan with fare comparisons and constraint checks.
- Verified, flagged, and unverified labels distinguish support, conflicts, and evidence gaps.

Constraint Validator

CONSTRAINT VALIDATOR	
Dietary	6 verified · 0 flagged · 0 unknown
Geographic	13 verified · 0 flagged · 1 unknown
Sankyu Halal Japanese Food Asakusa	Verified
RAMEN HONOLU SHINJUKUGYOENMAE	Verified
Café Restaurant Le Temps, Hotel Granvia Kyoto	Verified
Istanbul Saray	Verified
CHIBO Diversity Dotonbori	Verified
PHO ME HALAL RESTAURANT	Verified

PHO ME HALAL RESTAURANT	Verified
Senso-ji → Sankyu Halal Japanese Food Asakusa	Verified
Sankyu Halal Japanese Food Asakusa → Tokyo Skytree	Verified
Meiji Jingu → Shibuya Crossing	Verified
Shibuya Crossing → RAMEN HONOLU SHINJUKUGYOENMAE	Verified
RAMEN HONOLU SHINJUKUGYOENMAE → Shinjuku Gyoen National Garden	Verified
Café Restaurant Le Temps, Hotel Granvia Kyoto → Fushimi Inari Taisha	Verified
Arashiyama Bamboo Grove → Tenryu-ji	Verified

Tenryu-ji → Yasaka Shrine	Verified
Yasaka Shrine → Istanbul Saray	Verified
Osaka Castle → CHIBO Diversity Dotonbori	Unverified
CHIBO Diversity Dotonbori → Dotonbori Canal Walk	Verified
Namba Yasaka Shrine → PHO ME HALAL RESTAURANT	Verified
PHO ME HALAL RESTAURANT → Shinsekai	Verified
Kuromon Ichiba Market → Shinsaibashi-suji Shopping Street	Verified

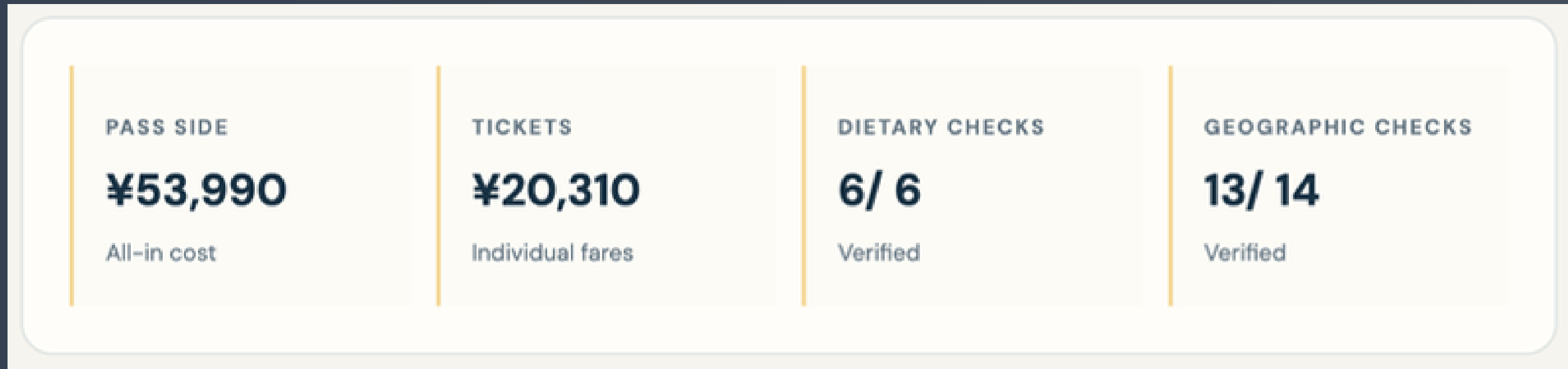
- Checks dining stops against curated dietary tags.
- Compares scheduled transfer gaps with curated travel-time estimates.
- Marks unsupported checks as unverified rather than assuming they pass.

Pass ROI Auditor



- Presents the calculator's BUY / DO NOT BUY recommendation.
- The second LLM explains the result without recomputing the figures.
- Separates numerical decision-making from natural-language explanation.

Calculator



- Compares individual ticket costs with the all-in pass cost for included segments.
- Accounts for applicable supplements and journeys not covered by the pass.
- Withholds the recommendation when required fare data is unavailable.

Scope: curated fare-bearing segments, not the total trip budget.

Evidence

EVIDENCE / GROUNDING

Point-to-point ticket total for this itinerary is 20310 yen.

Source: calculator:sumFares · Verified: Undated

Base price of "JR Pass 7-day Ordinary" is 50000 yen.

Source: jrpass-price-2026 · Verified: 2026-09-06

Narita Airport Terminal 1 to Keisei-Ueno by Keisei Skyliner costs 2580 yen.

Source: keisei-skyliner-fares-2026 · Verified: 2026-09-06

Tokyo to Kyoto by Hikari costs 13650 yen.

Source: smartex-reserved-fares-2025 · Verified: 2026-09-06

Kyoto to Shin-Osaka by Hikari costs 2670 yen.

Source: smartex-reserved-fares-2025 · Verified: 2026-09-06

Namba to Kansai Airport by Nankai Rapi:t Digital Ticket costs 1410 yen.

Source: nankai-rapit-digital-ticket-2026 · Verified: 2026-09-06

Narita Airport Terminal 1 to Keisei-Ueno by Keisei Skyliner is not covered by "JR Pass 7-day Ordinary" and adds 2580 yen.

Source: keisei-skyliner-fares-2026 · Verified: 2026-09-06

Namba to Kansai Airport by Nankai Rapi:t Digital Ticket is not covered by "JR Pass 7-day Ordinary" and adds 1410 yen.

Source: nankai-rapit-digital-ticket-2026 · Verified: 2026-09-06

- Shows fare claims with source identifiers and verification dates where available.
- Links the cost comparison to individual route segments and pass-coverage rules.
- Supports traceability without implying real-time fare updates.